

HW 9 Solutions

§2.1: 1-4, 8-11

§2.2: 1-6, 14 (explained)

§2.1

#1 In book.

#2. a) (a_n) (4, 5, 7, 11, 19, 35...)

$+2^n \rightarrow$ $+1$ $+2$ $+4$ $+8$ $+16$



These are Almost powers of 2.
from the common seq.

2^n : 1, 2, 4, 8, 16, 32

We want to turn 2^n into our
Sequence a_n by Adding 3

(Make 1 into 4)

$$a_n = 2^n + 3$$

check $i+1$!

$$a_0 = 2^0 + 3 = 1 + 3 = 4 \quad \checkmark$$

$$a_1 = 2^1 + 3 = 2 + 3 = 5 \quad \checkmark$$

$$a_2 = 2^2 + 3 = 4 + 3 = 7 \quad \checkmark$$

Good!

2b) $(a_n) = (0, 3, 8, 15, 24, 35)$

$\begin{array}{cccccc} & \nearrow & \nearrow & \nearrow & \nearrow & \nearrow \\ +3 & +5 & +7 & +9 & +11 \end{array}$

Almost perfect Squares.

$$n^2: 1, 4, 9, 16, 25,$$

$$a_n = n^2 - 1.$$

2c)

6, 12, 20, 30, 42, ...
↗ ↗ ↗ ↗
+6 +8 +10 +12

Notice Triangle #'s

$n =$ 1 2 3 4 5
 T_n 1, 3, 6, 10, 15, 21

$T_n \times 2 \rightarrow$ 2, 6, 12, 20, 30, 42

0 1 2 3 4
Start

$c_n \leftarrow$ index 0

$$c_n = 2T_{n+2} = 2 \cdot \frac{(n+2)((n+2)+1)}{2}$$

$$= (n+2)(n+3)$$

$$a_0 = 2 \cdot 3 = 6$$

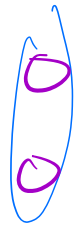
$$a_1 = 3 \cdot 4 = 12$$

$$a_2 = 4 \cdot 5 = 20 \quad \checkmark$$

This is the short version of what
I did in class.

2d) $s_n (0, 2, 7, 15, 26, 40, 57)$

House #3.



1
+



3
+

Triangle
Square

6
+

10
+

15
+

21
+

$$a_n = n^2 + \frac{n(n+1)}{2}$$

3a)

$$a_n = \frac{1}{2}(n^2 + n)$$

$$(a_n)_{n=0}^{\infty} = (0, 1, 3, 6, 10, \dots)$$

$$\begin{array}{ccccccc} & & & 1 & 2 & 3 & 4 \\ & & \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \\ & +1 & +2 & +3 & +4 & & \end{array}$$

Recursive

$$\begin{cases} a_n = a_{n-1} + n \\ a_0 = 0. \end{cases}$$

3b) $a_n = 2a_{n-1} - a_{n-2}, \quad a_0 = 0$
 $a_1 = 1$

$$(a_n)_{n=0}^{\infty} = (0, 1, 2, 3, 4, 5, 6, \dots)$$

$$\begin{array}{c} \nearrow \\ 2 \cdot 1 - 0 \\ \nearrow \\ 2 \cdot 2 - 1 \\ \nearrow \\ 2 \cdot 3 - 2 \\ \nearrow \end{array}$$

close to

$$\boxed{a_n = n}$$

* But why?

$$a_{n-1} = n-1$$

$$a_{n-2} = n-2$$

with Recursion formula.

$$a_n = 2a_{n-1} - a_{n-2}$$

$$= 2(n-1) - (n-2)$$

$$= 2n - 2 - n + 2$$

$$= n$$



3c)

$$a_n = n \cdot a_{n-1}, \quad a_0 = 1$$

$$(a_n)_{n=0}^{\infty} = (1, 1, 2, 6, 24, 120, \dots)$$

$$1, 2 \cdot 1, 3 \cdot 2 \cdot 1, 4 \cdot 3 \cdot 2 \cdot 1, 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$a_n = n!$$

4) In book.

8) In book.

$$9) \quad a_n = 3 \cdot 2^n + 7 \cdot 5^n$$

Show the Recursive formula works for a_n .

$$a_n = 7a_{n-1} - 10a_{n-2}$$

First find a_{n-1} and a_{n-2}

$$a_{n-1} = 3 \cdot 2^{n-1} + 7 \cdot 5^{n-1}$$

$$a_{n-2} = 3 \cdot 2^{n-2} + 7 \cdot 5^{n-2}$$

Then into the Recursive.

$$a_n = 7a_{n-1} - 10a_{n-2}$$

$$= 7 \cdot (3 \cdot 2^{n-1} + 7 \cdot 5^{n-1})$$

$$- 10 \cdot (3 \cdot 2^{n-2} + 7 \cdot 5^{n-2})$$

$$= 21 \cdot 2^{n-1} + 49 \cdot 5^{n-1} - 30 \cdot 2^{n-2} - 70 \cdot 5^{n-2}$$

Note $2^{n-2} = \frac{2}{2} \cdot 2^{n-2}$

$$= \frac{2 \cdot 2^{n-2}}{2}$$

$$= \frac{2^{n-1}}{2}$$

Similarly $5^{n-2} = \frac{5^{n-1}}{5}$

$$= 21 \cdot 2^{n-1} + 49 \cdot 5^{n-1} - 30 \cdot \frac{2^{n-1}}{2} - 70 \cdot \frac{5^{n-1}}{5}$$

$$= 21 \cdot 2^{n-1} + 49 \cdot 5^{n-1} - 15 \cdot 2^{n-1} - 14 \cdot 5^{n-1}$$

$$= 6 \cdot 2^{n-1} + 35 \cdot 5^{n-1}$$

$$= 6 \cdot \frac{2^n}{2} + 35 \cdot \frac{5^n}{5}$$

$$= 3 \cdot 2^n + 7 \cdot 5^n = a_n$$

$$a_0 = 10, a_1 = 41 \quad \text{Woooo!}$$

10.) Summ. id. c.

$$a_n = 2^n - 5^n$$

$$a_n = 7a_{n-1} - 10a_{n-2}$$

$$\begin{cases} a_{n-1} = 2^{n-1} - 5^{n-1} \\ a_{n-2} = 2^{n-2} - 5^{n-2} \end{cases}$$

$$a_n = 7(2^{n-1} - 5^{n-1}) - 10(2^{n-2} - 5^{n-2})$$

$$= 7 \cdot 2^{n-1} - 7 \cdot 5^{n-1} - 10 \cdot 2^{n-2} + 10 \cdot 5^{n-2}$$

$$= 7 \cdot 2^{n-1} - 7 \cdot 5^{n-1} - 10 \cdot \frac{2^{n-1}}{2} + 10 \cdot \frac{5^{n-1}}{5}$$

$$= 7 \cdot 2^{n-1} - 7 \cdot 5^{n-1} - 5 \cdot 2^{n-1} + 2 \cdot 5^{n-1}$$

$$= 2 \cdot 2^{n-1} - 5 \cdot 5^{n-1}$$

$$a_n = 2^n - 5^n \quad \checkmark$$

Wheee... Again.

$$a_0 = 0, \quad a_1 = 2 - 5 = -3$$

11) sum is 36.

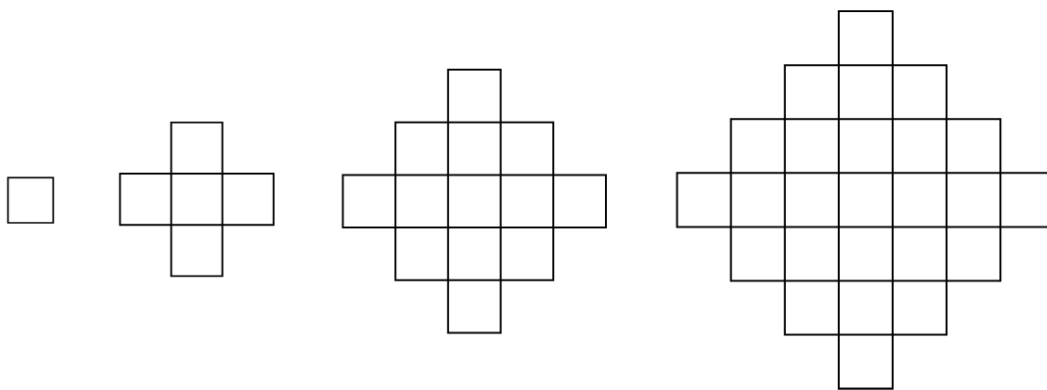
§2.2 #1-6, #14

#1-6 have solutions in Book.

#14



14. If you were to shade in a $n \times n$ square on graph paper, you could do it the boring way (with sides parallel to the edge of the paper) or the interesting way, as illustrated below:



The interesting thing here, is that a 3×3 square now has area 13. Our goal is the find a formula for the area of a $n \times n$ (diagonal) square.

- Write out the first few terms of the sequence of areas (assume $a_1 = 1$, $a_2 = 5$, etc). Is the sequence arithmetic or geometric? If not, is it the sequence of partial sums of an arithmetic or geometric sequence? Explain why your answer is correct, referring to the diagonal squares.
- Use your results from part (a) to find a closed formula for the sequence. Show your work. Note, while there are lots of ways to find a closed formula here, you should use partial sums specifically.
- Find the closed formula in as many other interesting ways as you can.

4)

$$a_1 = 1, \quad a_2 = 5, \quad a_3 = 13, \quad a_4 = 25$$

→

+4

$\times 4$

→

+8

$\times \frac{13}{5} ??$

→

+12

$\times \frac{25}{13} ???$

Not Arithmetic or geometric

It is Not going up by the same amount for Addition or Multiplication

It is going up by 4, 8, 12, 16

which looks to be Arithmetic.

so it is probably the sum of
an Arithmetic Seq.

$$1 = 1$$

$$5 = 1 + 4$$

$$13 = 1 + 4 + 8$$

$$25 = 1 + 4 + 8 + 12$$

b) We will find a closed
formula for the stuff
in the yellow box. Then
we will Add 1!

Our Almost Seq.

0, 4, 12, 24, 40?

$$\begin{array}{cccc} \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright \\ +4 & +6 & +12 & +16 \end{array}$$

Let's call c_n the stuff we are adding, call s_n the sum.

$$\begin{array}{llll} s_1 = 0 & \longrightarrow & c_1 = 4 & \downarrow +4 \\ s_2 = 4 & \longleftarrow & c_2 = 8 & \downarrow +4 \\ s_3 = 12 & \longleftarrow & c_3 = 12 & \downarrow +4 \\ s_4 = 24 & \longleftarrow & c_4 = 16 & \downarrow +4 \\ & & \vdots & \end{array}$$

Find a formula for c_n , then a formula for s_n .

c_n is arithmetic.

$$d = 4$$

$$\cancel{a = 4}$$

But it starts c_1 . If

there was a c_0 , it would be

$$c_0 = 0. \quad \leftarrow \text{red } a!!$$

$$C_n = 0 + 4 \cdot n$$

$$\boxed{C_n = 4n}$$

$S_1 = 0$
 $S_2 = \underline{C_1} = 4$
 $\underline{S_3} = \underline{C_1} + \underline{C_2} = 4 + 8 = 12$
 $S_4 = C_1 + C_2 + C_3 = 4 + 8 + 12 = 24$

Be careful about where to start!

$$\underline{S_4} = C_1 + C_2 + \underline{C_3} = 4 + 8 + 12 = 24$$

So

$$\boxed{\underline{S_n} = C_1 + C_2 + \dots + \underline{C_{n-1}}}$$

Gauss's Trick

$$\Rightarrow S_n = 4 + 8 + 12 + \dots + 4(n-1)$$

Reverse order

$$S_n = \underline{4(n-1)} + 4(n-2) + \dots + 4$$

$$\rightarrow 4 + 4(n-1) = 4 + 4n - 4 \\ = \boxed{4n}$$

Add Vertically

$$2s_n = (4n) + (4n) + (4n) \dots + (4n)$$

How Many terms??

Look back at s_n in the Box.

$$s_n = c_1 + c_2 + \dots + c_{n-1}$$

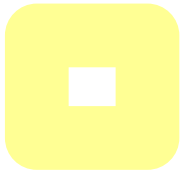
1 to $(n-1)$
is $(n-1)$ terms.

so

$$2s_n = 4n(n-1)$$

$$S_n = \frac{n^2 (n-1)}{2}$$

$$S_n = 2n(n-1)$$

And we have our Almost Seq.
 to get our original Seq.
 We need to add 1, like
 we said when we made the
 yellow Box .

$$a_n = S_n + 1$$

$$a_n = 2n(n-1) + 1$$

or

$$a_n = 2n^2 - 2n + 1$$

check!
Does it work?

$$a_1 = 2 \cdot 1^2 - 2 \cdot 1 + 1 = 1$$

$$a_2 = 2 \cdot 2^2 - 2 \cdot 2 + 1 = 8 - 4 + 1 = 5$$

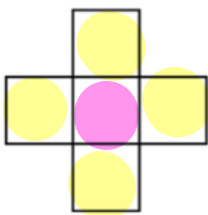
$$a_3 = 2 \cdot 3^2 - 2 \cdot 3 + 1 = 18 - 6 + 1 = 13$$

Looks good!

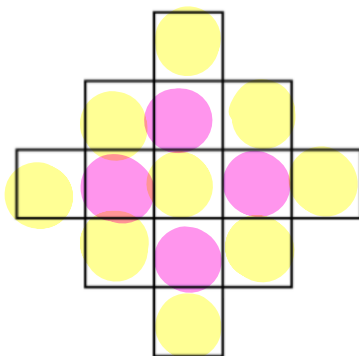
c) Fun other way!
OTZ you could have noticed...



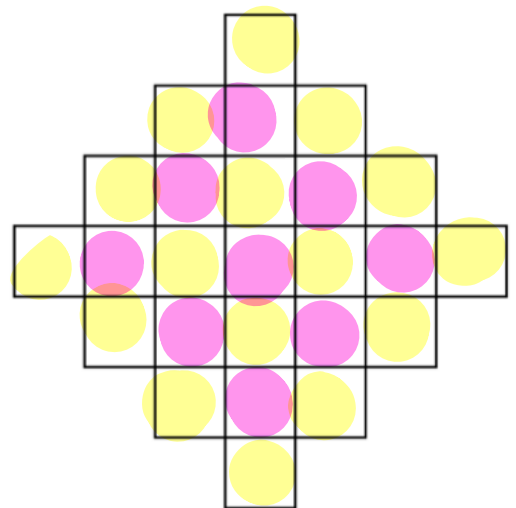
a_1



a_2



a_3



a_4

Notice the outside yellow dots for
example in G_3 form a
 3×3 normal.

And the inside purple dots
form a 2×2 square.

$$G_3 = 3 \cdot 3 + 2 \cdot 2 = 9 + 4 = 13$$

$$G_4 = 4^2 + 3^2 = 25$$

$$G_5 = 5^2 + 4^2 = 41$$

$$\Rightarrow \boxed{G_n = n^2 + (n-1)^2}$$